

Harbor Goods: migration to AWS

Fictional sample deliverable. The company Harbor Goods and all people, servers and data described here are invented. AWS documentation examples supply the account and network values. The repository files and the evidence regenerated by make evidence provide every number in this document.

Item	Value
Client	Harbor Goods, a mid-size online retailer (fictional)
Scope	12 servers in one data center; wave 1 in full, waves 2 and 3 at plan level
Target	AWS, us-east-1, one production account (123456789012)
Deliverables	Inventory, 7R classification, wave plan, Terraform, DMS tasks, cutover runbook, acceptance criteria
Revision	1.0

Executive summary

A four-hour Sunday window allows Harbor Goods to migrate its online store to AWS, with ordering unavailable to customers for no more than 30 minutes. During the seven days that follow, the store can move back to the data center at any time without losing orders.

The nine risks identified during discovery divide equally into high, medium and low severity; the complete register appears under "Risks." Had the three high-severity issues gone undetected, cutover would have failed.

Severity	Count	Examples
High	3	BI reads the replica that wave 1 retires; a table without a primary key; the payment provider allowlists the source IP
Medium	3	Nightly jobs inside the window; one-hour DNS TTL; PostgreSQL 14 to 16 during the move
Low	3	No WAF yet; application secret without rotation; DMS TLS without certificate checks

The three priority recommendations are:

1. Request the payment provider's allowlisting of the new egress addresses now (prerequisite P-04). This prerequisite takes the longest to arrange, and go-live cannot proceed without it.
2. Approve the proposed first wave, moving the storefront and orders API to ECS Fargate and replatforming the database onto RDS for PostgreSQL 16 Multi-AZ. Use AWS DMS for full load followed by change data capture, with a reverse task supporting rollback.
3. Complete the full rehearsal (P-09); keep the scheduled date only if the freeze takes no more than 27 minutes (G-04).

Scope and method

`docs/methodology.md` defines the sequence: discovery, 7 Rs classification, dependency-based waves, sizing from measurements, data migration planning, a runbook expressed as data and a target built as code. Automated checks cover every step. E-06 records the plan checks, E-07 records the runbook checks, and E-08 contains the Terraform test results.

Application code changes and a cost estimate are excluded; a real engagement would include an AWS Pricing Calculator export. Detailed runbooks for waves 2 and 3 are also outside this scope and will be written after wave 1 exits.

Current state

The inventory (E-01) lists eight applications across twelve servers. An HAProxy pair fronts two storefront servers and two orders API servers. PostgreSQL 14 runs on a primary and streaming replica, holding 182 GB across 55 tables in five schemas. The remaining servers host nightly batch jobs, a self-hosted BI tool, supplier SFTP and a rarely used intranet wiki. Both the storefront and API tolerate a 30-minute write freeze and require recovery within 30 minutes (RTO), with zero data loss (RPO 0).

7R classification

Strategy	Servers	Why
Replatform	SRV-01 to SRV-07	Stateless web and API tiers run as containers unchanged; a managed load balancer and database remove hand patching
Retire	SRV-08, SRV-12	Multi-AZ replaces the replica; the wiki has no edits in the discovery sample
Rehost	SRV-09	Batch jobs depend on local paths; containerizing them needs code changes
Repurchase	SRV-10	14 dashboards move to a managed BI service
Retain	SRV-11	Supplier contracts allowlist the SFTP address

For each server, `plan/classification.yaml` and E-02 record the selected target, the reason for choosing it and the alternative rejected. Refactoring is not assigned to any server because no business driver justifies that rework as part of the migration (ADR 0001).

Wave plan

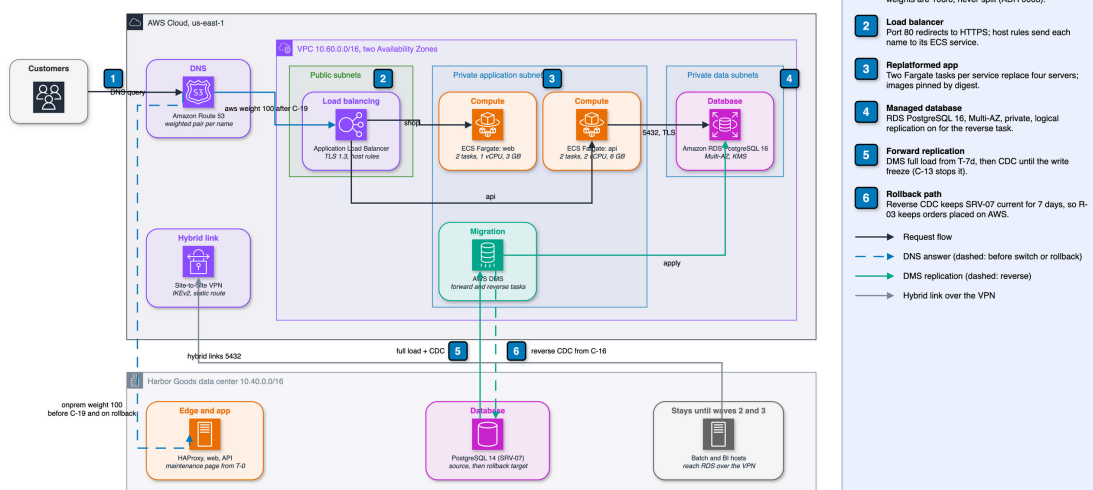
Wave	Servers	Exit criterion
1 Commerce core	SRV-01 to SRV-08	All acceptance criteria met after seven days of hypercare
2 Batch and intranet	SRV-09, SRV-12	Three nightly runs match the on-premises row counts
3 Reporting	SRV-10	Finance signs off the rebuilt dashboards after one month-end close
Retained	SRV-11	Revisit when supplier contracts renew

Between waves, three connections use the Site-to-Site VPN (E-03). After wave 1, batch and BI connect to RDS; after wave 2, batch connects to the retained SFTP server. Database access through Terraform is limited to the batch and BI hosts, using the /32 addresses specified in the plan.

Target architecture

Harbor Goods wave 1: target architecture and cutover path

Route 53 weighted records move customers from the data center to ECS and RDS; AWS DMS keeps both databases in step (fictional)



Customer traffic follows Route 53 weighted records to the selected destination. Using a TLS 1.3 policy, an Application Load Balancer directs each host name to its ECS Fargate service in private subnets. The RDS for PostgreSQL 16 database uses Multi-AZ and a customer managed KMS key for encryption; its data subnets have no internet route. From the application subnets, AWS DMS connects to the data center through the VPN and to RDS within the VPC.

Discovery p95 measurements determine sizing, with 25 percent headroom (E-04). Each orders API task has 2 vCPU and 6 GB, and each storefront task has 1 vCPU and 3 GB. The database uses a db.r7g.2xlarge instance and 400 GB of gp3 storage.

Data migration

At T-7d, the forward task begins a full load of all five schemas, then streams changes up to the write freeze (E-05). DMS cannot keep audit.request_log synchronized because the table

lacks a primary key and its retention job deletes rows. A `pg_dump` export transfers its history at T-24h; `\copy` transfers the remaining rows during the freeze. Manual steps M-01 to M-05 handle sequences, roles and grants, which DMS does not copy. When cutover occurs, a reverse task begins streaming changes from RDS to the data center, allowing rollback without data loss after go-live (ADR 0002).

Cutover and rollback

The runbook defines 13 prerequisites and 23 timed steps, with 8 go/no-go criteria distributed across three checkpoints. It also includes 7 validation queries, 7 rollback triggers and 3 rollback procedures:

Phase	Window	Rollback	Time to roll back
Preparation	T-72h to T-0	R-01: abort, nothing to undo	15 min
Write freeze	T-0 to T+27m	R-02: restart the data center application; RDS is reloaded before the next attempt	15 min
Hypercare	T+27m to T+7d	R-03: maintenance page on AWS, drain the reverse task, flip DNS back	25 min

For both DNS names, the switch sets Route 53 weights to 0/100 from 100/0. Because the database permits only one writer, the Terraform module rejects intermediate weights (ADR 0003). Three days before the switch, the TTL is reduced from 3600 seconds to 60 seconds. Fixture-based execution on PostgreSQL 14 and 16 verified the validation queries (`make sql-check`).

Acceptance criteria

Acceptance of wave 1 requires seven days of hypercare and satisfaction of all ten conditions in `runbooks/acceptance-criteria.md`. These include a 99.9 percent pass rate for synthetic checks, orders API p95 no higher than 350 ms, an actual write freeze of no more than 30 minutes, zero lost orders and completion of a point-in-time restore within 60 minutes.

Risks

ID	Severity	Risk	Evidence	Treatment
RISK-01	High	BI dashboards read the replica that wave 1 retires	E-03	Repoint BI to RDS over the VPN in wave 1 (C-20)
RISK-02	High	<code>audit.request_log</code> has no primary key and a delete job	E-05	Exclude from DMS; export the history and copy the tail in the freeze (M-05)
RISK-03	High	The payment provider allowlists the data center egress address	E-01	Register the NAT gateway addresses before cutover (P-04)
RISK-04	Medium	JOB-01 and JOB-03 run inside the window against the database	E-07	Disable at T-80m (C-06), final dump at C-07, re-enable batch after the switch
RISK-05	Medium	One-hour TTL on the public names	E-07	Lower to 60 s at T-72h (C-01)
RISK-06	Medium	PostgreSQL 14 to 16 during the move	E-04	Full rehearsal on 16 (P-09); smoke tests before the switch
RISK-07	Low	No AWS WAF in front of the load balancer (none on premises either)	E-08	Add managed rules in count mode during hypercare, then block
RISK-08	Low	The application database secret has no rotation	E-08	Enable rotation once the API reloads credentials, after hypercare
RISK-09	Low	DMS endpoints use TLS without certificate verification	E-08	Import the RDS and data center CAs and switch to <code>verify-full</code>

Limits

- All material is fictional and synthetic, so none of the findings apply to an actual system.
- Offline testing establishes consistency among the plan, runbook and Terraform, and confirms Terraform's intended properties. It does not establish that AWS accepts every setting. The manually run `make test-live` checks the network and database slice in a sandbox account.
- Planning for waves 2 and 3 covers dependencies; after wave 1 exits, their runbooks will be developed using the same format.

Evidence index

ID	File	Produced by
E-01	Inventory summary	scripts/build_evidence.py
E-02	Classification and waves	scripts/build_evidence.py
E-03	Dependencies by wave	scripts/build_evidence.py
E-04	Target sizing	scripts/build_evidence.py
E-05	DMS coverage	scripts/build_evidence.py
E-06	Plan checks	scripts/check_plan.py rules
E-07	Runbook checks	scripts/check_runbook.py rules
E-08	Terraform tests	terraform test with mock providers

Fictional sample prepared for a portfolio. Harbor Goods does not exist.